Soloman Northrop

Lead Software Engineer | Full-Stack, Robotics and AI Platforms | C#/.NET, TypeScript, Flutter, Unity

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SUMMARY

Lead software engineer, seven years in. Built and shipped complete production platforms as the only engineer, front end and back end: a Flutter robotics control platform now running on a commercial security robot, a WebXR physical-therapy platform in use at 15+ clinics, and a multi-tenant cloud platform for streaming and AR/VR. Main languages are C#/.NET and TypeScript. Also ship in Flutter, Unity and Python. Have put LLMs, on-device computer vision and physics simulation into products customers use. Own architecture through release and work directly with founders, clinicians and hardware engineers.

SKILLS

Languages: C#, TypeScript/JavaScript, Dart, Python, SQL

Backend and data: ASP.NET Core, Entity Framework, Node.js, Express, Colyseus, PostgreSQL, SQL Server, Supabase, REST APIs, webhooks, multi-tenant architecture, Hangfire

Front end and apps: React, Vite, Svelte, Three.js, Flutter, Unity, WebXR, Expo/React Native, Electron, Razor

Cloud and delivery: AWS (ECS Fargate, ALB, EFS, ECR, SES), Fly.io, Cloudflare, Auth0, Docker, GitHub Actions, CI with automated tests (node:test, Playwright, flutter_test), multi-platform release builds

AI and robotics: LLM integration (OpenAI including realtime voice, Gemini, Claude), agent orchestration and MCP, on-device inference (ONNX Runtime, TFLite, TensorRT on Jetson), YOLO, depth and segmentation deployment, ZED stereo vision, GNSS/RTK, mapping and navigation, physics simulation (Rapier, WebGPU), serial and firmware protocols

EXPERIENCE

Lead Software Engineer (full-time) | Robocor — Apr 2024 – Present

Remote, Florida

- Designed and built Robocor 2.0, a Flutter/Dart robotics control platform targeting iOS, Android, macOS, Windows, Linux and web, with automated release pipelines for Windows, macOS and Jetson/Linux. It handles autonomous navigation, telemetry, computer vision, fleet management and remote operation on a Supabase backend with multi-tenant worker services on Fly.io. Sole software engineer, reporting to the founder and working with the hardware and firmware team.
- Shipped the Guardian security robot to a live customer site (Nimnicht Chevrolet). Wrote the whole software stack, from the firmware serial protocol and motor watchdog up to the cloud backend and operator app.
- Built the on-device perception pipeline: YOLO26 and YOLO-World detection, Depth Anything v2 and UniDepth metric depth, and SegFormer segmentation, exported to ONNX Runtime, TFLite and TensorRT on NVIDIA Jetson, with a ZED stereo camera and a Flutter ONNX Runtime plugin I maintain. Fine-tuned a custom segmentation model as a field experiment.
- Added GNSS/RTK path following, offline mapping and navigation, and an OpenAI GPT-4o operator agent with realtime voice so staff can talk to the robot.
- Created corBench, a three.js/WebGPU and Rapier physics test bench with virtual sensor boards driving the real parsers unchanged. 580+ recorded runs used to reproduce field failures and validate navigation changes before they reach hardware.

Software Developer III, Full Stack | VEERA Medical — Apr 2023 – Present

Hybrid, Florida

- Built and launched MedXR (Recovr360), a white-label WebXR physical-therapy platform. The patient wears a Meta Quest headset; the practitioner works from a browser. In use at more than 15 clinics for home and in-clinic therapy. Live at app.recovr360.com.
- Wrote all of it: the Node/TypeScript real-time backend (Colyseus rooms with a strict session lifecycle, Express), the Svelte and Three.js WebXR client, the Unity scene exporter and native Quest client, the patient progress pipeline and clinic monitoring.

- Migrated the platform from Fly.io to AWS ECS Fargate behind an ALB with EFS, ECR and Cloudflare, and rolled out Auth0 PKCE login behind feature flags.
- Added accessible patient input: Xbox Adaptive Controller, wheelchair joystick HID adapters and standard gamepads.
- Presented the platform live at ACRM 2023. A practicing clinician ran the wireless demo.

Software Developer II | VEERA Medical Oct 2021 – Mar 2023

- Built the first VEERA Medical cross-platform application on the Reach Cloud codebase I co-authored, in C#, Unity, JavaScript and ASP.NET Core: Razor UIs, authentication and the clinician data pipeline.

Co-Founder and Lead Engineer (side venture) | Reach Cloud Jan 2022 – Present

Remote, St. Petersburg, FL

- Built and maintain a multi-tenant cloud platform for streaming, remote device management and cross-platform multiplayer VR/AR/MR, with a Unity SDK and public API. Live at reachcloud.org.
- Wrote the developer docs and the C# generator that builds them. Set the engineering roadmap and shipped across platforms.

Junior Software Application Programmer (part-time) | Cardno May 2019 – Sep 2023

Clearwater, FL

- Maintained and extended the Georgia Department of Transportation permit system in C# and SQL. Added batch reporting and export and Hangfire background processing, and made the site faster.

INDEPENDENT PRODUCTS

- Solla Code (t3.codes): independently maintained open-source fork of T3 Code, extended into a control surface for AI coding agents. Multi-agent orchestration, MCP tools, scheduled autonomous tasks, shared browser automation, approval gating and usage guards. 850+ commits and 500k+ lines of my own since July 2026 across the Electron, web and Expo mobile clients; nightly prereleases and tagged stable releases.
- SkyCal (skycalapp.com): family calendar and planning SaaS in React, TypeScript and Vite on Supabase. Stripe household billing, Google and Outlook calendar sync (Microsoft Graph), a GPT realtime voice assistant and a referral program; 50+ test files including Stripe sandbox tests.
- Glow Music Therapy platform (glowmt.com): public site, family portal and practice-management backend in framework-free Node and PostgreSQL with WebAuthn passkeys, RBAC, an append-only audit trail, Stripe invoicing and HIPAA-aligned deployment gates. In production for a paying practice.
- GenEdit Pro: open-source browser-first AI video editor on ffmpeg.wasm (React, Zustand, Playwright tests).

EDUCATION

St. Petersburg College | Associate of Science, Computer Programming | 2016 – 2017